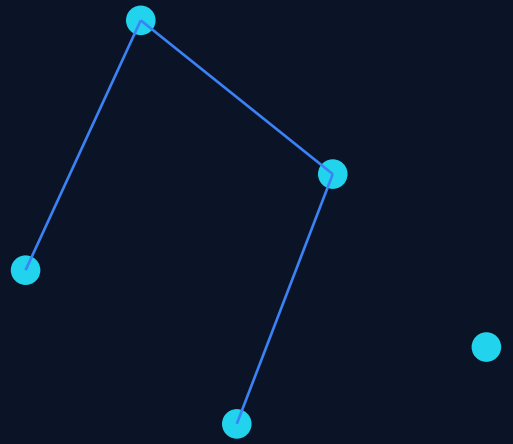
AI-BASED SMART GARBAGE DETECTION AND MONITORING ROBOT

with Live Video, GPS, and Long-Range Communication

AI • Computer Vision • Robotics • Embedded Systems • IoT



ESP32-CAM • YOLO • GPS
UART • Wi-Fi • LoRa (planned)

Engineering project presentation | Proposed architecture, development in progress

A INTRODUCTION

Smart waste management: why it matters

Traditional

Fixed collection routes and manual inspection. Littered areas are found late, if at all.

Smart

Sensors, cameras and AI detect waste and report where it is, so effort goes where needed.

This project

A mobile robot that sees garbage, classifies it and attaches a GPS position to each detection.

A INTRODUCTION

Problem statement

Slow discovery

Manual patrols miss areas and find waste late.

No location data

Reports rarely say exactly where garbage was seen.

No structured record

Without logs, patterns and hotspots cannot be analysed.

Labour intensive

Repetitive inspection ties up people who could collect instead.

Motivation and significance

Environmental

Faster detection means faster cleanup and less litter spreading.

Technical

Combines AI vision, embedded control and GPS on low-cost hardware.

Practical

Location-tagged records can guide cleanup crews or future collector robots.

Existing approaches vs proposed solution

Approach	Strength	Limitation
Manual patrol	Flexible, human judgement	Slow, no data record
Fixed cameras	Continuous watch	Covers one location only
Smart bins	Fill-level alerts	Do not find loose litter
Proposed robot	Mobile, AI detection, GPS tagging	Prototype in development; GPS and range limits

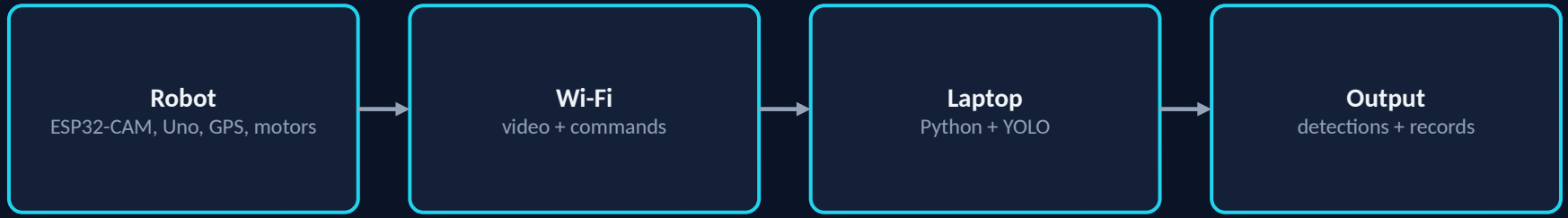
Project objectives



Planned extension: long-range LoRa telemetry for status and detection events.

B PROPOSED SYSTEM

Project overview



Eyes

ESP32-CAM streams live video.

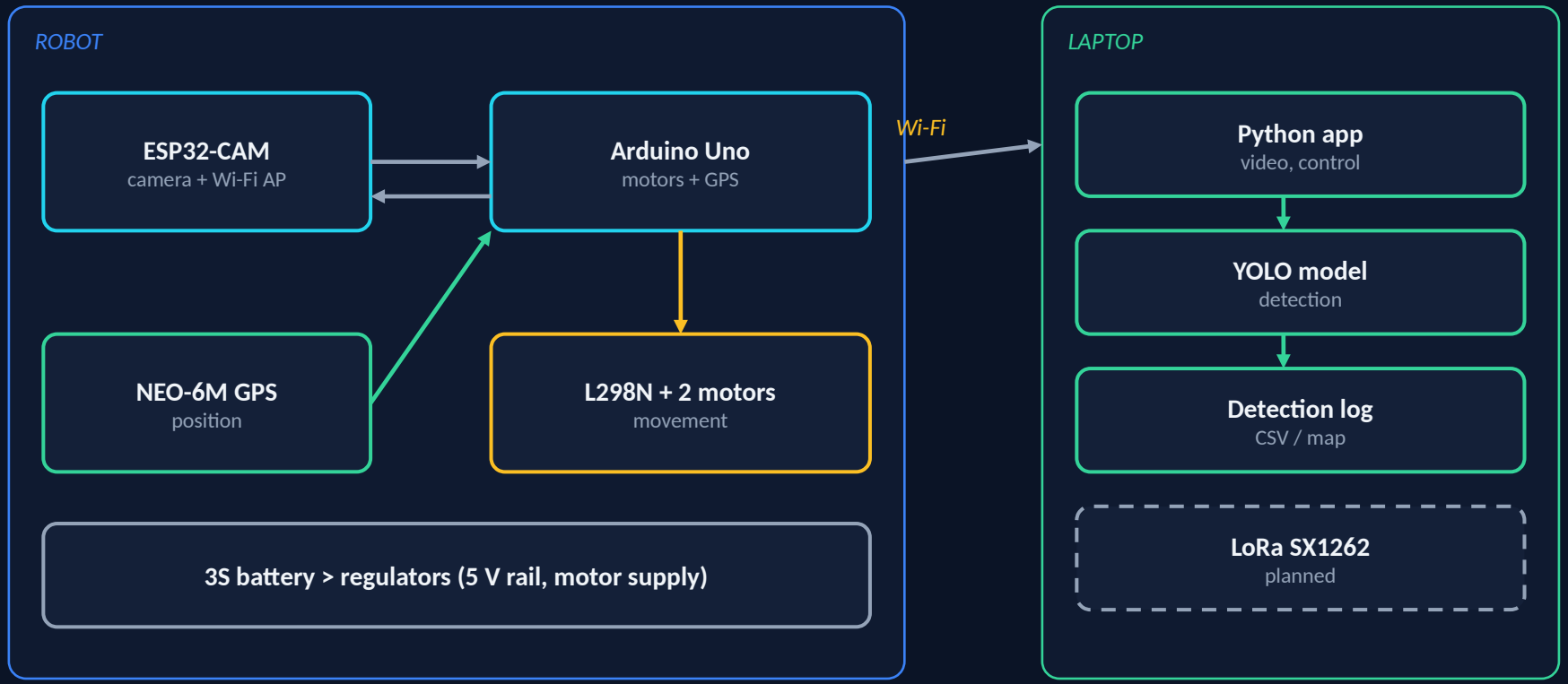
Brain

Laptop runs AI detection.

Body

Uno, L298N and motors move the robot; GPS gives position.

System architecture



Complete working principle



Intended workflow. Safety stop on lost video is part of the design.

Hardware and software integration

Link	Between	Carries
Wi-Fi (access point)	Laptop and ESP32-CAM	Video stream, commands
UART serial	ESP32-CAM and Uno	Drive and speed commands, GPS text
UART (software serial)	NEO-6M and Uno	NMEA position sentences
PWM + direction pins	Uno and L298N	Motor speed and direction
LoRa (planned)	Robot and base	Telemetry, detections

Hardware components overview

Component	Purpose
ESP32-CAM AI-Thinker	Camera capture and Wi-Fi video streaming
Arduino Uno	Motor control and GPS interfacing
NEO-6M GPS	Geographical positioning
L298N driver + 2 DC motors	Robot movement
Robot chassis, 3S lithium battery	Platform and power
Laptop	Video processing and AI inference
LoRa SX1262 (proposed)	Long-range communication extension

ESP32-CAM: camera and live streaming

What and why

Low-cost camera + Wi-Fi board; lets the robot send video without cables.

How it works

Initialises the camera, starts a Wi-Fi access point named GarbageBot (intended 192.168.4.1), streams JPEG frames.

Role and links

Video to laptop over Wi-Fi; UART to the Uno. Needs a stable 5 V supply with headroom for current spikes.

Arduino Uno and motor control

Direction

IN1 to IN4 set forward, reverse and turn for the two motors.

Speed (PWM)

Enable pins carry PWM; duty cycle sets average motor power.

Safety and GPS

Designed to stop when commands cease; reads NEO-6M via software serial.

C HARDWARE

NEO-6M GPS module

What it gives

Latitude, longitude, time and fix status as NMEA text.

Needs

Open-sky view, antenna facing up, away from motors and Wi-Fi radio.

Role

Position attached to each detection record. Accuracy to be verified in testing.

L298N motor driver and DC motors



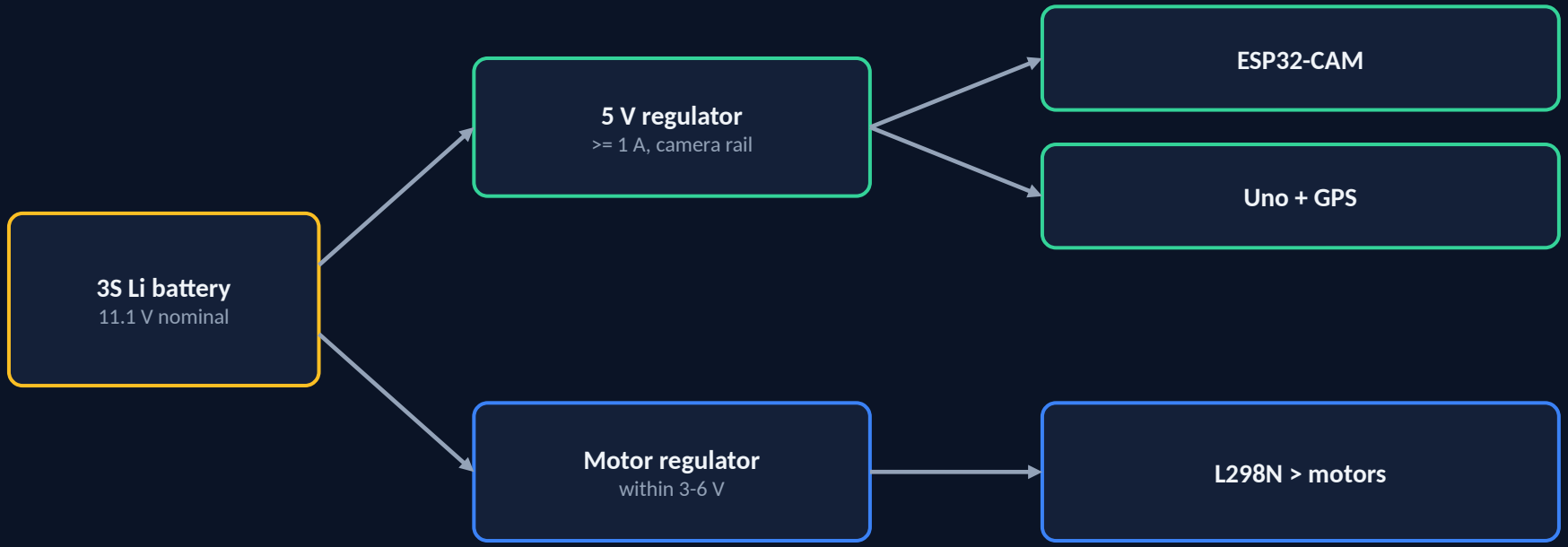
Wiring idea

ENA/ENB take PWM; IN1-IN4 set direction. Remove the ENA/ENB jumpers so PWM works.

Caution

Do not feed raw 11.1 V to motors rated 3-6 V. Share a common ground.

Battery, power distribution and chassis



Common ground across every module. Add a capacitor near the ESP32-CAM supply to absorb spikes.

Software architecture

ESP32-CAM firmware

Camera init, Wi-Fi access point, video stream.

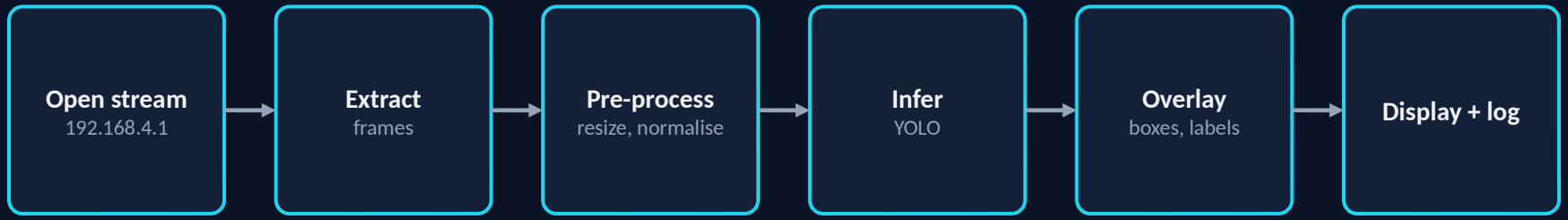
Arduino firmware

Motor direction, PWM speed, GPS parsing, serial link.

Python application

Stream receive, frames, YOLO inference, overlay, GPS association, logging.

Live video processing workflow



Frame rate and latency are to be measured; none are claimed here.

YOLO-based garbage detection

Object detection

Finds objects and their location in an image.

YOLO idea

One forward pass predicts boxes, classes and confidence: fast.

Outputs

Box, class, confidence score; optional segmentation mask for shape and size.

Status

Intended YOLO-based approach. Model version and training data not fixed.

AI inference pipeline

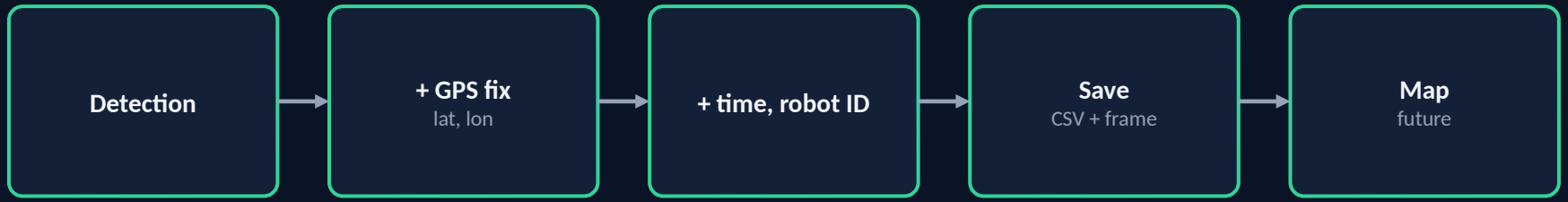


Illustrative pipeline; thresholds will be tuned during testing.

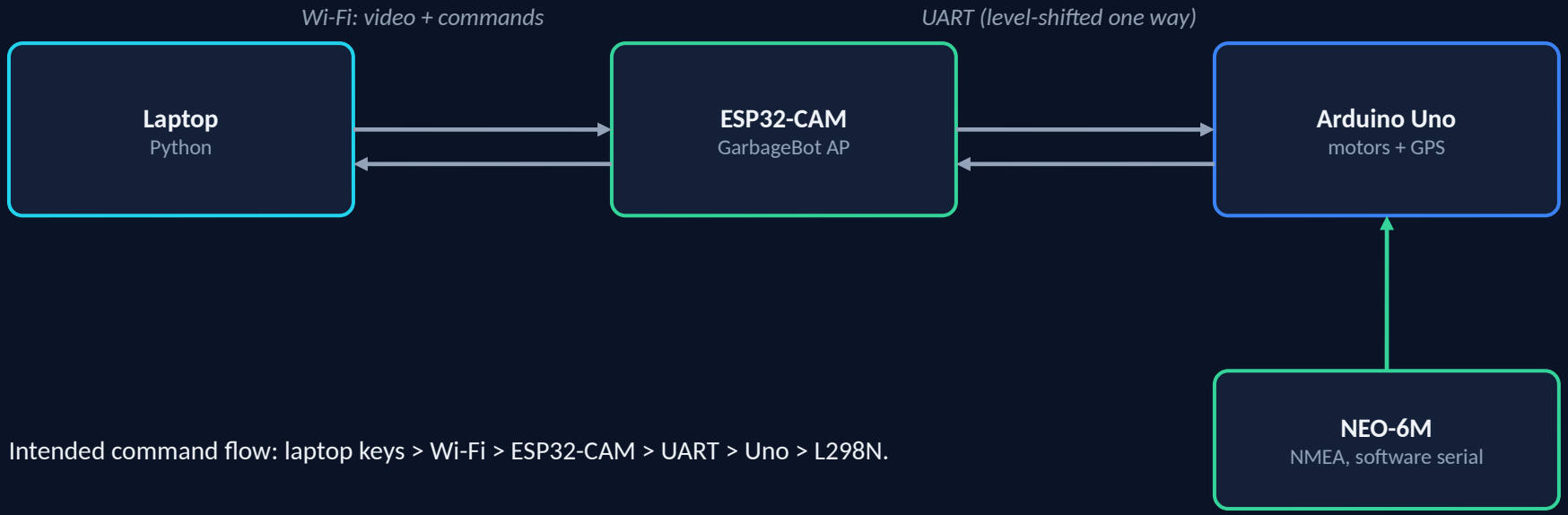
Detection records and data management

Time	Type	Conf.	Lat	Lon	Robot	Frame
10:42:07	plastic bottle	0.81	26.0000	80.0000	R1	img_001.jpg
10:42:31	paper	0.74	26.0001	80.0001	R1	img_002.jpg

ILLUSTRATIVE SAMPLE ONLY: not real measurements.



Wi-Fi and serial communication



Intended command flow: laptop keys > Wi-Fi > ESP32-CAM > UART > Uno > L298N.

Proposed LoRa / SX1262 long-range link

Planned role

Send GPS, robot status, battery level and detection events beyond Wi-Fi range.

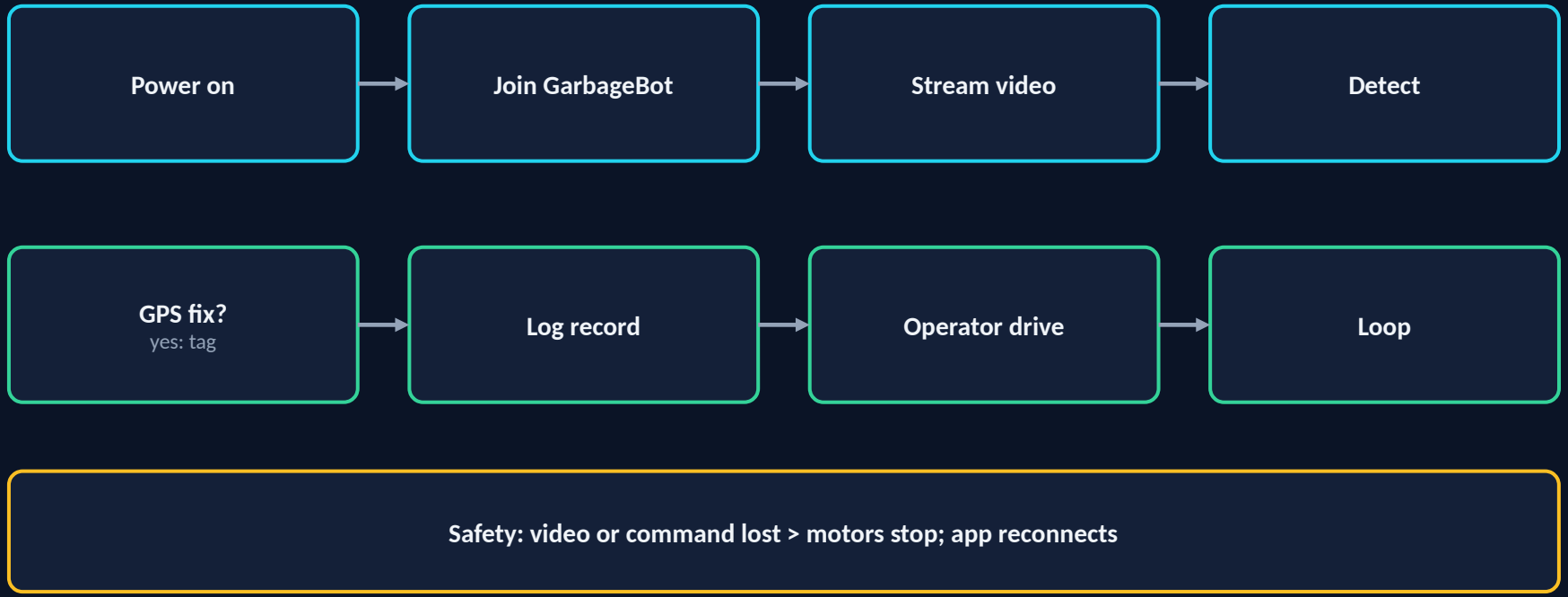
Not for video

Low data rate suits small telemetry packets only.

Status

PLANNED EXTENSION. Not implemented or tested.

End-to-end system workflow



F EVALUATION

Expected outcomes and testing methodology

AI

Precision, recall and confidence behaviour on labelled test images.

Streaming

Frame rate, latency and dropouts at different distances.

GPS

Fix time and position error against known reference points.

Robot

Control response, battery runtime and safety stop reliability.

F EVALUATION

Technical challenges and limitations

Power

Camera brownouts from supply dips; needs regulated 5 V and filtering.

GPS

Weak signal indoors; metre-level error; robot position, not item position.

Range

Wi-Fi limited; LoRa planned for telemetry.

AI

Accuracy depends on dataset and lighting; to be validated.

F EVALUATION

Future scope

AI

Custom garbage model, size estimation.

Autonomy

Patrol, obstacle avoidance.

Mapping

Map of garbage points shared with collector robots.

Hardware

Pickup arm, battery monitor, LoRa link.

Conclusion and Q&A

Summary

Proposed robot that sees garbage with YOLO, locates it with GPS and records it; LoRa planned.

Questions?

Thank you.